

Presentation of a paper:

A Test for Relative Efficiency and Application to Indian Agriculture
Lawrence J. Lau & Pan A. Yotopoulos (1971)

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AEEM - FEBA - Sofia University St. Kliment Ohridski - coh. 2026/27

April 17, 2026

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Background of the authors

Academic

Lawrence J. Lau

is a Hong Kong economist, former Chancellor of Chinese University of Hong Kong & economics professor at Stanford. He has a B.S. in physics & economics from Stanford, M.A. & Ph.D. from Berkeley. In 1966 after graduating the M.A. he was joined the Stanford Faculty, after the paper in 1976 he became a Professor of Economics in 1976.

Pan A. Yotopoulos

was a Greek Economist in the field of economic development and agricultural economics, economic demography, international trade, production and consumption theory. He joined Stanford as a Associate Professor in 1968.

Connection of the paper with my experience

My bachelor degree thesis

Analysis of Profitability in Small Individual Grape Producers in North Macedonia — Published in Macedonian in the UGD Economics Faculty Yearbook 2025 (Vol 27 No.1)

Using a 4-plantation sample, I collected all economic inputs and outputs to run scenario analyses and evaluate their effects on:

- H0 - Lack of Capital → higher operating costs → financial unsustainability
- H1 - Short - term subsidies → lower product quality → spreading of Phytopathogens
- H2 - Small producers lack economies of scale for sustainable practices

Quick conclusion

The government's subsidy policy aims to preserve the status quo and protect political capital.

North Macedonia Grape & India

Investment management ✂ Micro 3

Grape



India



The crucial theory

Back to Micro 3

Starting of with the most basic formula:

$$V^1 = A^1 F(X^1); \quad V^2 = A^2 F(X^2)$$

Technical efficiency if $A^1 > A^2$ produces better profits.

Same conclusion was derived with the improvement in the process in the North Macedonian small sample case - Cost reduction

After all the derivation the authors present the "Unit-Output-Price" & profit function.

$$P = \frac{P'}{p} = F(X_1, \dots, X_m; Z_1, \dots, Z_n) - \sum_{i=1}^m c_i X_i$$

$$X_i^* = f_i(c, Z), \quad i = 1, \dots, m$$

The crucial theory 2

Continuation from the first theory slide

Continuation from the last slide:

$$\Pi = p \left[F(X_1^*, \dots, X_m^*, Z_1, \dots, Z_n - \sum_{i=1}^m c_i X_i^*) \right] = G(p, c'_1, \dots, c'_m; Z_1, \dots, Z_n)$$

So the final UOP profit function is:

$$\Pi^* = \frac{\Pi}{p} = G^*(c_1, \dots, c_m; Z_1, \dots, Z_n)$$

Using Shepar-Uzawa-McFadden Lemma equation:

$$V^* = \Pi^*(c, Z) - \sum_{i=1}^m \frac{\delta \Pi^*(c, Z)}{\delta c_i} \cdot c_i$$

The crucial theory 3

Cobb–Douglas derivation

After all the derivation we get to:

$$\ln \Pi_a^1 = \ln A_*^1 + \sum_{j=1}^n \alpha_j^* \ln c_j^1 + \sum_{j=1}^n \beta_j^* \ln Z_j^1$$

$$\ln \Pi_a^2 = \ln A_*^1 + \ln \frac{A_*^2}{A_*^1} + \sum_{j=1}^n \alpha_j^* \ln c_j^2 + \sum_{j=1}^n \beta_j^* \ln Z_j^2$$

Where:

- Π_a - the profit/production aggregate
- A_* - the productivity
- $\ln \frac{A_*^2}{A_*^1}$ - change in productivity
- c_j - conventional input
- Z_j - quasi-fixed exogenous factor Z_j

Empirical Application

Empirical Cobb–Douglas applied

$$\ln \prod_a^1 = \ln A_*^1 + \alpha_1^* \ln w + \beta_1^* \ln K + \beta_2^* \ln T$$

$$\ln \prod_a^2 = \ln A_*^1 + \ln \frac{A_*^2}{A_*^1} + \alpha_1^* \ln w + \beta_1^* \ln K + \beta_2^* \ln T$$

Final estimation:

$$\ln \prod = \alpha + S + \sum_{i=1}^4 \delta_i^* D_i + \alpha_1^* \ln w' + \beta_1^* \ln K + \beta_3^* \ln T$$

Results

From the Indian Sample Case

TABLE 3—COMPARISON OF THE RATES OF RETURN ON
FIXED CAPITAL AND LAND BETWEEN
LARGE AND SMALL FARMS

	Large Farms	Small Farms
Geometric Means		
$\bar{\Pi}$ (rupees)	2,184.62	493.90
$\bar{K}$ (rupees)	51.35	22.89
$\bar{T}$ (acres)	23.81	3.99
Rates of Return		
$\frac{\partial \Pi}{\partial K}$ (rupees per rupee)	-25.02	-12.69
$\frac{\partial \Pi}{\partial T}$ (rupees per acre)	164.88	222.44

Source: *Farm Management Studies.*

Ending remarks

I am delighted for your attention.

Q & A?